

HANDS-ON · BEGINNER FRIENDLY

AI & Machine Learning for Civil Engineers.

From engineering data to **prediction**, **explainability** & **uncertainty**

08

LIVE SESSIONS

90 min

PER SESSION

12 hrs

LIVE CODING



100% Google Colab — zero installs

No setup. Runs in any browser — even on a Chromebook.



No prior Python or ML needed

We start from engineering problems, not equations.



Live code-along, real datasets

Concrete, geotech, water & transport data.



Notebooks, assignments & solutions

Yours to keep and revisit after the course.

YOUR INSTRUCTOR

Dr. Abdul Waris

Research Associate, University of Siegen, Germany

Ph.D., IIT Hyderabad · Prime Minister's Research Fellow

🏆 Research Excellence Award 2026

🌐 Talks in USA · Norway · Spain

The 8-session journey →

PROBLEM > DATA > MODEL > VALIDATE > INTERPRET > DECIDE

S01

Python for Civil Engineers

Colab · NumPy · pandas

S02

Raw data to ML-ready

EDA · outliers · leakage

S03

Your first prediction model

Regression · R^2 · validation

S04

Trees & ensembles

Random Forest · boosting

S05

Classification for decisions

Precision · recall · F1

S06

Artificial neural networks

Neurons to predictions

S07

Explainable AI & judgement

SHAP · physical plausibility

S08

Uncertainty & reliability

Monte Carlo · P(failure)

WHO IT'S FOR

B.Tech / M.Tech students · PhD scholars · Faculty · Practising engineers

📜 Certificate 📺 Session recordings 🚫 Limited seats

Register now



qudelastudios.com · Dates announced soon